

B. Tech 3rd Semester Sessional Examination II
Mathematics To Civil Engineering (BMA23211T)
Marks – 20, Time –1 Hour

Answer the following questions

1. Choose the correct options:

1X6=5

- a) The period of the function $f(x) = \sin x$ is
 - (i) π
 - (ii) 2π
 - (iii) 3π
 - (iv) 4π
- b) The function $f(x) = \cos x, 0 < x < \pi$ is
 - (i) an odd function
 - (ii) an even function
 - (iii) neither an odd nor an even function
 - (iv) both an odd and even function
- c) The Fourier series of even function contains only
 - (i) sine terms
 - (ii) cosine terms
 - (iii) both sine and cosine terms
 - (iv) none of these
- d) The degree of the partial differential equation $\left(\frac{\partial z}{\partial x}\right)^4 + \frac{\partial^3 z}{\partial y^3} = 2 \frac{\partial z}{\partial x}$ is
 - (i) 1
 - (ii) 2
 - (iii) 3
 - (iv) 4
- e) The one-dimensional Wave equation is given by
 - (i) $\frac{\partial y}{\partial x} = c^2 \frac{\partial^2 y}{\partial t^2}$
 - (ii) $\frac{\partial y}{\partial t} = c^2 \frac{\partial^2 y}{\partial x^2}$
 - (iii) $\frac{\partial^2 y}{\partial x^2} = c^2 \frac{\partial^2 y}{\partial t^2}$
 - (iv) $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$

2. a) Find a Fourier Series of the function $f(x) = x, -\pi \leq x \leq \pi$.

Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$

5+2=7

- b) Solve the one-dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ subject to boundary conditions $u(0, x) = 0 = u(l, t)$ and initial condition $u(x, 0) = u_0$

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